AI Playbook and Toolkit for Public Health Departments

AI, Health Literacy, Plain Language, and Accessibility

HED 110

This module prepares public health staff to use AI responsibly when developing health education materials, training content, handouts, scripts, scenarios, and learner supports. It focuses on health literacy, plain language, accessibility, cultural relevance, and human review. Because individual agencies approve different tools, the module does not recommend a specific AI platform. Instead, it emphasizes awareness of approved tools, prohibited data, review steps, and safe drafting practices.

Level	introductory/applied
Primary track	Health Education Track
Appears in tracks	health-education

Learning Objectives

- Explain how AI can support health literacy, plain-language review, accessibility, and audience adaptation.
- Identify risks when AI-generated education materials are inaccurate, stigmatizing, culturally inappropriate, inaccessible, or inconsistent with approved guidance.
- Apply a structured review process before using AI-assisted educational content with staff, partners, clients, patients, or community members.
- Create a plain-language and accessibility review checklist for an AI-assisted public health education product.

Module Overview

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Jurisdiction and Agency Policy Note

This national-level training module is intended for public health education, planning, and workforce development. It does not replace agency policy, legal review, privacy review, security review, procurement review, communications clearance, civil rights review, Tribal consultation, or governance approval. Learners should confirm applicable requirements, approved tools, data rules, and review pathways within their own jurisdiction and organization before implementing AI-supported workflows.

Why This Topic Matters for Public Health AI

Health education materials often influence what people understand, remember, trust, and do. AI can help staff draft materials quickly, simplify complex language, generate examples, and adapt content for different audiences, but speed can create risk when content is not carefully reviewed. A draft that sounds clear may still be medically inaccurate, culturally inappropriate, inaccessible to people with disabilities, or inconsistent with agency guidance. Public health agencies need staff who can use AI as a drafting aid while preserving accuracy, equity, accessibility, and trust.

Health literacy is not only about reading level. It includes whether the material explains the action people should take, why the action matters, where to get help, and how to apply the guidance in real life. AI tools may simplify language but remove nuance, omit eligibility limits, overgeneralize recommendations, or fail to explain local context. Human review is therefore essential before AI-assisted materials are used in programs, trainings, outreach, or public communication.

Public Health Example

A local health department is preparing a handout for families about preventing heat-related illness during a summer heat event. A health educator uses an approved AI drafting tool to create a first draft at a sixth-grade reading level and asks for versions for older adults, outdoor workers, and caregivers of young children. The health educator then checks the content against approved health guidance, removes advice that does not match local resources, adds phone numbers for cooling centers, and reviews the handout for language access and accessibility. The final product is not considered AI-approved simply because it was generated quickly; it becomes usable only after subject matter, plain-language, accessibility, and program review.

Technical, Programmatic, and Operational Deep Dive

A responsible workflow begins before the prompt is written. Staff should identify the purpose of the educational material, the audience, the approved source material, the intended reading level, and the review process. They should also determine whether the AI tool is approved for the task and whether any sensitive data, partner information, participant stories, or non-public guidance must be excluded. This preparation keeps the AI task focused and reduces the chance that staff will paste inappropriate information into an unapproved environment.

The prompt should direct the AI tool to use only approved source text or general language support, not to invent public health facts. When AI is used for readability, translation support, or audience adaptation, staff should compare the output to the original source to ensure meaning has not changed. Accessibility review should include headings, reading order, alt text needs, color contrast, screen-reader compatibility, and whether instructions rely only on visual cues. Cultural review should consider whether examples, tone, imagery, assumptions, and calls to action fit the intended audience.

Agencies should treat AI-assisted health education content as draft material. Drafts should go through the same or stronger review process as materials developed without AI, especially when the content involves clinical guidance, emergency instructions, eligibility, legal requirements, or community-specific concerns. Staff should document the source material used, the reviewers involved, and any major edits made after AI drafting. This documentation supports transparency and helps teams learn which AI uses are helpful and which create too much review burden.

Risks, Failure Modes, and Guardrails

A common failure mode is assuming that easier language is automatically more accurate or more equitable. AI may reduce reading level while removing details that certain groups need to act safely. It may also create examples that unintentionally stigmatize people, oversimplify risk, or imply that individuals are responsible for barriers created by systems. Guardrails include using approved sources, requiring human review, involving audience representatives when feasible, and checking whether materials explain practical actions without blame or assumptions.

Another risk is using AI to translate or localize content without qualified review. Automated translation can produce fluent but incorrect language, omit culturally important context, or use terms that do not fit the community. Staff should use AI translation only within agency-approved workflows and should involve qualified bilingual reviewers for public-facing materials. When qualified review is not available, the material should not be treated as ready for public use.

A third risk is entering sensitive stories, participant feedback, or program records into an AI tool to make examples more realistic. Even de-identified examples can sometimes reveal identities in small communities or specialized programs. Staff should use fictional, composite, or generalized examples unless the agency has approved the tool and data use. Practical safeguards include no-sensitive-data warnings, source tracking, reviewer checklists, and documentation of final approval.

Application to AI-Supported Workflows

In AI-supported workflows, this module helps staff build a review pathway for educational materials before they are used in training, outreach, or public-facing programs. The workflow should identify who drafts, who reviews accuracy, who reviews accessibility, who reviews cultural relevance, and who gives final clearance. The AI tool should never be the final reviewer or the source of authority for health guidance. Staff should also identify when a material must be routed to communications, legal, clinical, program, equity, accessibility, or language-access reviewers.

Reflection Questions

What types of health education content in your organization might benefit from AI-assisted drafting or plain-language review?

What information should never be entered into an AI tool when creating health education materials?

Who should review AI-assisted materials before they are used with the public, learners, or program participants?

Practical Exercise

Select a public health education topic such as heat safety, immunization reminders, lead exposure prevention, food safety, or asthma trigger reduction. Draft a one-page plain-language review checklist for an AI-assisted handout. Include the audience, approved source material, reading-level goal, accessibility checks, cultural relevance checks, subject matter reviewers, and final approval steps.

Example: For a heat safety handout for older adults, the checklist might require that the draft uses approved heat guidance, explains symptoms that require urgent help, names local cooling resources, avoids blaming people who cannot afford air conditioning, uses large readable text, includes alt text for icons, and receives review from emergency preparedness, health education, and communications staff before release.

Expected Artifact or Evidence

Plain-language and accessibility review checklist for one AI-assisted public health education product.

Brief note identifying the approved source material and required reviewers.

One example of a revision made to improve accuracy, accessibility, or audience fit.

Knowledge Check

- 1. Which statement best describes responsible use of AI for public health education materials?
- 2. What is one reason AI-generated plain language can still be risky?
- 3. Which information should staff avoid entering into an unapproved AI tool?
- 4. Who should provide final approval for AI-assisted educational materials?
- 5. What should be included in a review checklist?

References and Resources

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